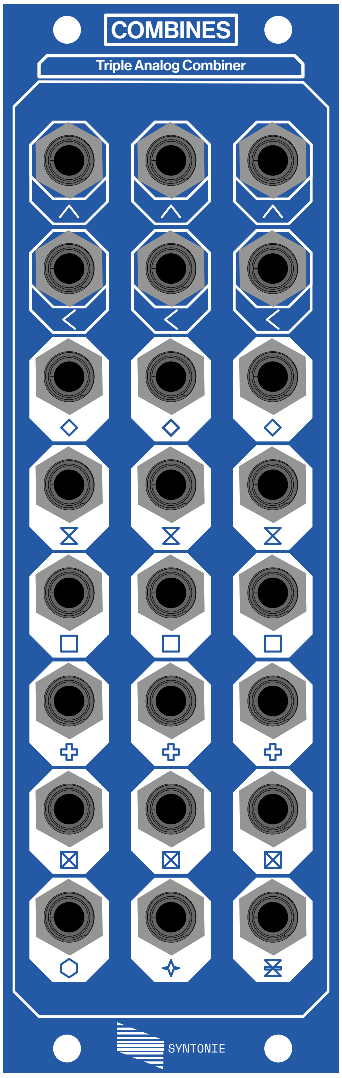
# Combines



**Triple Analog Combiner ¬**
User documentation

SYNTONIE




Combines is triple analog combiner, allowing to combine two signals in many different ways: average, difference, minimum, maximum, analog XOR, etc...

Each channel is identical at the exception of the last output, which is a mix of three of the previous outputs, different on each channel.

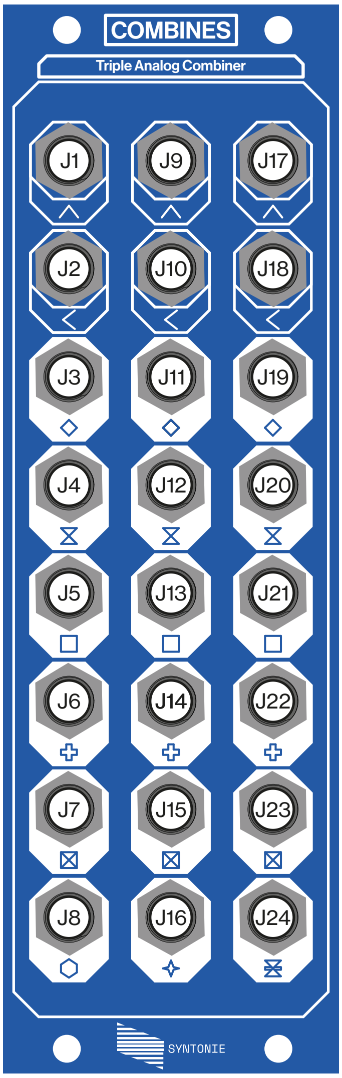
## Specifications

– 8HP
– 125 mA +12V (16pin or DC)
– 0 mA -12V
– 0 mA +5V
– 42mm depth

Special thanks to: **Yves Usson** for the Mini-Maxi Voltage Processor,, from which the analog logic section is derived. **Lorenzo Ferronato** for the documentation design // And of course, **everyone who has supported Syntonie until now & those who will support it in the future.**

## Interface






**(J1)** Input A channel 1 : 0-1V, 100kΩ, minijack

**(J2)** Input B channel 1: 0-1V, 100kΩ, minijack

**(J3)** Average channel 1: 0-1V,  75Ω, minijack

**(J4)** Difference channel 1: 0-1V, 75Ω, minijack

**(J5)** Minimum channel 1: 0-1V, 75Ω, minijack

**(J6)** Maximum channel 1: 0-1V, 75Ω, minijack

**(J7)** Analog XOR channel 1: 0-1V, 75Ω, minijack

**(J8)** Mix channel 1: 0-1V, 75Ω, minijack

**(J9)** Input A channel 2 : 0-1V, 100kΩ, minijack

**(J10)** Input B channel 2: 0-1V, 100kΩ, minijack

**(J11)** Average channel 2: 0-1V,  75Ω, minijack

**(J12)** Difference channel 2: 0-1V, 75Ω, minijack

**(J13)** Minimum channel 2: 0-1V, 75Ω, minijack

**(J14)** Maximum channel 2: 0-1V, 75Ω, minijack

**(J15)** Analog XOR channel 2: 0-1V, 75Ω, minijack

**(J16)** Mix channel 2: 0-1V, 75Ω, minijack

**(J17)** Input A channel 3: 0-1V, 100kΩ, minijack

**(J18)** Input B channel 3: 0-1V, 100kΩ, minijack

**(J19)** Average channel 3: 0-1V,  75Ω, minijack

**(J20)** Difference channel 3: 0-1V, 75Ω, minijack

**(J21)** Minimum channel 3: 0-1V, 75Ω, minijack

**(J22)** Maximum channel 3: 0-1V, 75Ω, minijack

**(J23)** Analog XOR channel 3: 0-1V, 75Ω, minijack

**(J24)** Mix channel 3: 0-1V, 75Ω, minijack

**Overview:**

Each channel of Combines features the following combinations:

- AVG: average of the two inputs, (A+B)/2
- DIFF: difference of the two inputs, ((A-B)/2)+0.5)
- MIN: minimum of the two inputs, min(A, B)
- MAX: maximum of the two inputs, max(A, B)
- XOR: absolute of the difference of the two inputs, abs(A-B)
- MIX: mix of three of the above, scaled to 1V, different on each of the three channels
    - CH1: (AVG+DIFF+MIN)/3
    - CH2: (AVG+MIN+XOR)/3
    - CH3: (AVG+DIFF+XOR)/3

As the front panel suggests, Combines can be used to combine a horizontal and vertical triangle signals (like rectified ramps or oscillators), resulting in 2D shapes, represented on the outputs. Of course, the inputs are not limited to triangle waves, as any signals can be combined together, such as different wave shapes, as well as 2D shapes to create even more complex shapes or an external video signal.

It features 3 channels to allow for RGB processing, however the channels can be chained together as well, increasing the complexity of the resulting output.

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